Analysis of the Chaotic Regime in the Filtered Hénon Map: Effect of FIR Filter Type and Design Window

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***Abstract*—The insertion of finite impulse response (FIR) filters in the feedback loop of chaotic generators is a known strategy for bandwidth limitation in chaos-based communication systems (CBCS). Although the filter does not prevent master-slave synchronization, its interaction with the nonlinearity of the generating map may destroy the chaotic behavior, so the presence of chaos must be verified for each design. This work investigates, by means of numerical simulations, the conditions under which chaos is preserved in the two-dimensional Hénon map with internal FIR filters, considering four filter types (lowpass, highpass, bandpass, and bandstop) and seven design windows, including a sweep of the Kaiser shape parameter. Points of the order-cutoff plane whose orbits become unbounded are classified as divergent. The remaining points are classified as chaotic when the largest Lyapunov exponent, which measures the average exponential rate at which nearby orbits separate, is positive, and as periodic otherwise. The results show that filtering does not eliminate the chaotic regime but reorganizes it in the parameter space in a way that depends on both the filter type and the design window. Therefore, the spectral specification of the filter alone does not ensure a chaotic generator, and the dynamics must be verified for each design.**

***Index Terms*—Chaos-based communication systems, chaotic synchronization, Hénon map, FIR filters, window design, Lyapunov exponent.**

I. Introduction

DETERMINISTIC discrete-time chaotic signals are bounded in amplitude, aperiodic, and exhibit sensitive dependence on initial conditions (SDIC), so that orbits starting arbitrarily close separate, on average exponentially fast [1]. They also have a broadband spectrum and a noise-like appearance, which motivates their use as carriers in communication systems [2]. Among the advantages usually attributed to such carriers are the difficulty of interception by unauthorized receivers, which places them among the physical-layer security approaches [3], the spreading of the transmitted spectrum, and, for some modulation schemes such as differential chaos shift keying (DCSK), robustness to multipath fading [4]. This interest grew after Pecora and Carroll [5] showed that two identical chaotic systems can be synchronized despite SDIC, when the receiver is driven by state variables of the transmitter.

In chaos-based communication systems (CBCS), the message is embedded in a chaotic carrier at the transmitter and recovered at the receiver. In coherent CBCS, the receiver synchronizes a replica of the transmitter generator with the received signal and uses it to extract the message. Non-coherent architectures, in contrast, recover the information from statistical or structural properties of the received signal without a synchronized replica [4]. Examples of CBCS include chaotic modulation schemes [2], image coding and encryption [6], systems based on three-dimensional attractors [7], and phase-locked loops for systems based on chaotic attractors [8].

The broadband spectrum of chaotic signals, however, conflicts with the fact that practical channels are band-limited. If the spectrum of the transmitted signal is not confined to the channel passband, the distortion introduced by the channel prevents the receiver from synchronizing, and the message is lost [9]. To overcome this limitation, Eisencraft *et al.* [10] proposed inserting a discrete-time linear filter in the feedback loop of the chaotic generator, and Fontes and Eisencraft [11] showed that the filter does not prevent synchronization, a result already indicated in [10]. Further practical limitations, such as finite-precision digital implementation, were investigated by Borges and Eisencraft [12]. The lowpass case corresponds to a baseband channel, but other spectral masks arise in practice. Highpass filtering removes the DC component of the chaotic signal, which carries no information and is blocked by AC-coupled channels. Bandpass filtering confines the signal to an allocated band, and bandstop filtering creates a spectral notch to avoid narrowband interference or to coexist with other systems [13]. From a dynamical standpoint, these four types also span the range of DC gains G of the filter, which determines the fixed point of the expanded system [14].

Inserting a filter in the feedback loop, however, changes the dynamics of the generator, and nothing guarantees that the transmitted signal remains chaotic. Once the unfiltered pair synchronizes, the filter preserves synchronization for any choice of coefficients [15], but it may destroy the chaotic

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behavior of the generator. Borges and Eisencraft showed numerically that the coefficients determine whether the generated signals are chaotic, periodic, or divergent, for a two-coefficient filter [16] and for lowpass filters of increasing order [17], with bifurcation cascades, coexisting attractors, crises, and shrimp-shaped periodic islands [18]. Borges *et al.* extended this analysis to finite impulse response (FIR) filters with more coefficients and different zero placements [14] and to second-order all-pole filters [19]. The preservation of chaos must therefore be verified for each filter design. It remains open which combinations of filter type, design window, order, and cutoff frequency keep the generator chaotic, and this is the question addressed here.

Previous studies varied the coefficients of a two-coefficient filter [16], the zero placement of the FIR filter [14], and the order and cutoff frequency of lowpass filters designed with the Hamming and Blackman windows [14], [17]. In practice, however, an FIR filter is specified by its frequency response and designed by a method such as windowing, so the filter type and the design window are natural design choices. This work investigates how these two choices affect the preservation of chaos in the Hénon map with an internal FIR filter. Lowpass, highpass, bandpass, and bandstop filters are designed with seven windows, including a sweep of the Kaiser shape parameter [20], and for each configuration the filter order and the cutoff frequency are varied to build maps that classify the generated signals as chaotic, periodic, or divergent. The parameters of the Hénon map and the bandwidth of the bandpass and bandstop filters are kept fixed.

The remainder of this paper is organized as follows. Section II presents the preliminary concepts: the Hénon map, the Hénon map with an FIR filter in the feedback loop, the largest Lyapunov exponent, the FIR design windows, and the synchronization argument that motivates the study. Section III describes the methodology, Section IV presents and discusses the results, and Section V summarizes the conclusions, the contributions of the work, and future directions.

II. Preliminary Concepts

This section introduces the concepts needed to follow the rest of the paper. The Hénon map is presented first and then extended with an FIR filter in the feedback loop. The largest Lyapunov exponent, used to classify the signals generated by both systems, comes next, followed by the windows used to design the filter and by the synchronization argument that motivates the study.

A. Hénon Map

Consider the Hénon map [21]

$$\begin{cases} x_1(n+1) = \alpha - x_1^2(n) + \beta\, x_2(n), \\ x_2(n+1) = x_1(n), \end{cases} \tag{1}$$

where $n \in \{0, 1, 2, \ldots\}$ is the discrete time, $x_1(n)$ and $x_2(n)$ are the state variables, and α and β are real parameters. Grouping the state variables in the vector $\mathbf{x}(n) = [x_1(n)\ \ x_2(n)]^T$, where $(\cdot)^T$ denotes transposition, (1) can be written compactly as $\mathbf{x}(n+1) = \mathbf{F}(\mathbf{x}(n))$. The sequence of states $\{\mathbf{x}(n)\}_{n\geq 0}$ obtained by iterating $\mathbf{F}$ from an initial condition $\mathbf{x}(0)$ is called an *orbit* [1].

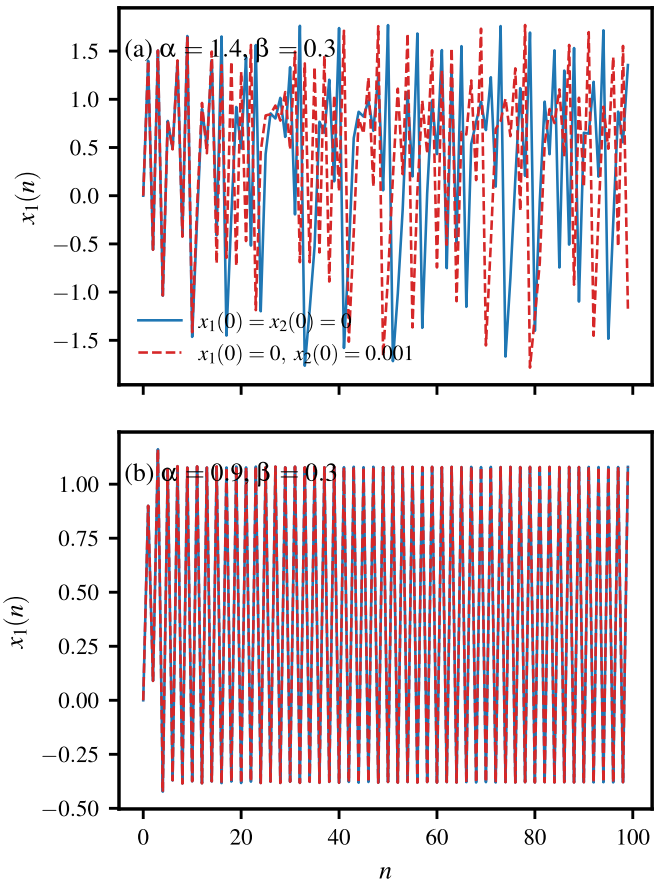


Fig. 1. Segments of $x_1(n)$ obtained from (1) with $x_1(0) = x_2(0) = 0$ (solid blue) and $x_1(0) = 0$, $x_2(0) = 0.001$ (dashed red). (a) $\alpha = 1.4$, $\beta = 0.3$: chaotic signal. (b) $\alpha = 0.9$, $\beta = 0.3$: periodic signal (period 2).

Depending on the parameters and on the initial condition, an orbit may converge to a fixed point, converge to a periodic orbit, behave chaotically, or diverge [1]. A *fixed point* is a state $\mathbf{x}^\star$ left unchanged by one iteration, $\mathbf{x}^\star = \mathbf{F}(\mathbf{x}^\star)$, so that an orbit started at $\mathbf{x}^\star$ stays there for all n. An orbit is *bounded* when there is a finite M such that $\|\mathbf{x}(n)\| \leq M$ for all n, and *divergent* otherwise [1]. For (1), the condition $\mathbf{x}^\star = \mathbf{F}(\mathbf{x}^\star)$ gives two fixed points, of which the positive one is used as the reference state in this work.

Fig. 1 illustrates the chaotic and the periodic behaviors with segments of $x_1(n)$ generated from two initial conditions that differ by 10^{-3} in $x_2(0)$, as in [17]. In Fig. 1(a), with $\alpha = 1.4$ and $\beta = 0.3$, the signal is aperiodic and, although the two initial conditions are nearly indistinguishable, the distance between them first exceeds 0.1 at $n = 11$ and reaches order one at $n = 18$; this is the SDIC of chaotic signals. In Fig. 1(b), with $\alpha = 0.9$ and $\beta = 0.3$, both orbits converge to the same period-2 orbit, which alternates between $x_1 \approx 1.08$ and $x_1 \approx -0.38$, and the difference between the initial conditions is lost. Throughout this work, $\alpha = 1.4$ and $\beta = 0.3$ are adopted; for these classical values, (1) has a strange attractor of fractal dimension ≈ 1.26 [22].

B. Hénon Map With an FIR Filter in the Feedback Loop

This subsection defines the filter, the expanded system obtained when the filter is inserted in the map, and the quantities of that system used in the rest of the paper. To limit the bandwidth of the generated signal, Eisencraft *et al.* [10] and Fontes and Eisencraft [11] proposed filtering $x_1(n)$ with a finite impulse response (FIR) filter and feeding the filter output back into the map. The output of the filter is

$$x_3(n) = \sum_{j=0}^{N_s-1} c_j\, x_1(n-j), \tag{2}$$

where c_j, $0 \le j \le N_s - 1$, are the N_s filter coefficients. The filter order is $N_z = N_s - 1$, equal to the number of zeros of the transfer function obtained by taking the z-transform of (2) with zero initial conditions [23],

$$H(z) = \frac{X_3(z)}{X_1(z)} = \sum_{j=0}^{N_s-1} c_j\, z^{-j}. \tag{3}$$

Evaluating $H(z)$ on the unit circle, $z = e^{j\omega}$, gives the frequency response $H(e^{j\omega})$, where $\omega \in [0, \pi]$ is the normalized angular frequency in rad/sample. The frequency response fixes the passband, the transition width, and the stopband attenuation delivered by a given set of coefficients [23]. In this work, it is specified by the filter type (lowpass, highpass, bandpass, or bandstop) and by a cutoff frequency ω_c. Its value at $\omega = 0$ is the DC gain of the filter, $G = H(e^{j0}) = \sum_j c_j$ [23].

The filtered signal $x_3(n)$ replaces $x_1(n)$ in the nonlinear term of (1). Because (2) depends on the past samples $x_1(n-1), \ldots, x_1(n-N_s+1)$, the pair (x_1, x_2) is no longer a complete state of the system. These samples are therefore stored as auxiliary state variables $x_4, \ldots, x_{N_s}$, each holding a one-step delay of the preceding one. For $N_s \ge 3$, the resulting expanded system is [14], [17]

$$\begin{cases} x_1(n+1) = \alpha - x_3^2(n) + \beta\, x_2(n), \\ x_2(n+1) = x_1(n), \\ x_3(n+1) = c_0\big(\alpha - x_3^2(n) + \beta\, x_2(n)\big) + c_1\, x_1(n) \\ \qquad\qquad + c_2\, x_2(n) + c_3\, x_4(n) + \cdots + c_{N_s-1}\, x_{N_s}(n), \\ x_4(n+1) = x_2(n), \\ x_j(n+1) = x_{j-1}(n), \quad j = 5, \ldots, N_s, \end{cases} \tag{4}$$

whose dimension is $K = N_s$; for $N_s = 3$, the terms in $x_4, \ldots, x_{N_s}$ are absent. For $N_s \le 2$, the filter output $x_3(n) = c_0 x_1(n) + c_1 x_2(n)$ (with $c_1 = 0$ for $N_s = 1$) is a function of the current state, so no auxiliary variable is needed and the system keeps dimension $K = 2$:

$$\begin{cases} x_1(n+1) = \alpha - \big(c_0 x_1(n) + c_1 x_2(n)\big)^2 + \beta\, x_2(n), \\ x_2(n+1) = x_1(n). \end{cases} \tag{5}$$

For $N_s = 1$ and $c_0 = 1$, the filter is the identity and the original map (1) is recovered. Equations (4) and (5) are the forms implemented in the package [24].

As for (1), the fixed points of (4) are the solutions of $\mathbf{x}^\star = \mathbf{F}(\mathbf{x}^\star)$, where $\mathbf{F}$ now denotes the right-hand side of (4). At a fixed point all delayed samples are equal, $x_1 = x_2 = x_4 = \cdots = x_{N_s} = p$, and $x_3 = Gp$, which gives $G^2p^2 + (1-\beta)p - \alpha = 0$ [14], [15]. Its positive root is written here as

$$p^+ = \frac{2\alpha}{(1-\beta) + \sqrt{(1-\beta)^2 + 4\alpha G^2}}, \tag{6}$$

which is algebraically equal to the usual form $\big[-(1-\beta) + \sqrt{(1-\beta)^2 + 4\alpha G^2}\big]/(2G^2)$ but avoids the cancellation of that form when $G \to 0$, as in highpass and bandpass filters, for which (6) tends to $\alpha/(1-\beta)$. The positive fixed point is $\mathbf{p}^+ = [p^+ \;\; p^+ \;\; Gp^+ \;\; p^+ \cdots p^+]^T$ (or $[p^+ \;\; p^+]^T$ for $N_s \le 2$); it depends on the filter only through its DC gain G and is used to initialize the simulations (Section III-B).

C. Largest Lyapunov Exponent

The sensitivity of a system to its initial conditions can be quantified by how fast two very close states move apart as they evolve. Lyapunov proposed measuring this separation under the assumption that it grows exponentially, so that the Lyapunov exponent is the natural logarithm of the average rate at which two nearby orbits separate [1], [17]. An exponent equal to $\ln 2$ along a given direction, for instance, means that the distance between the orbits doubles, on average, at each iteration in that direction, whereas a negative exponent means that the orbits approach each other. The SDIC of a chaotic signal is therefore measured by the Lyapunov exponent: a computationally generated signal that is bounded, aperiodic, and has at least one positive Lyapunov exponent is chaotic [1].

A map of dimension K has K Lyapunov exponents, one for each direction of the state space, denoted in decreasing order by $\lambda_1 \ge \lambda_2 \ge \cdots \ge \lambda_K$. They are computed from the *Jacobian matrix* $\mathbf{J}(n)$, the $K \times K$ matrix of partial derivatives $[\mathbf{J}(n)]_{ik} = \partial F_i / \partial x_k$ evaluated at $\mathbf{x}(n)$, which describes how a small perturbation of the state is transformed in one iteration [1]. For the expanded system (4), it is [14], [17]

$$\mathbf{J}(n) = \begin{pmatrix} 0 & \beta & -2x_3(n) & 0 & \cdots & 0 \\ 1 & 0 & 0 & 0 & \cdots & 0 \\ c_1 & c_0\beta + c_2 & -2c_0 x_3(n) & c_3 & \cdots & c_{N_s-1} \\ 0 & 1 & 0 & 0 & \cdots & 0 \\ \vdots & & & \ddots & & \vdots \\ 0 & \cdots & 0 & 0 & 1 & 0 \end{pmatrix}, \tag{7}$$

in which only the third row depends on the filter coefficients; for $N_s \le 2$, $\mathbf{J}(n)$ is the 2×2 Jacobian of (5). For the map (1), $\det \mathbf{J}(n) = -\beta$ for every n, so its two exponents satisfy $\lambda_1 + \lambda_2 = \ln|\beta|$ [1], an identity used in Section IV-A to validate the implementation.

The exponents are computed by the tangent-map method [1], [17]. A hypersphere of unit radius centered at $\mathbf{x}(0)$ is represented by a matrix $\mathbf{U}$ whose columns are its orthogonal axes. Each iteration of the map turns the hypersphere into an ellipsoid, and after n iterations the axes of the ellipsoid are the columns of $\mathbf{J}(n-1)\cdots\mathbf{J}(1)\mathbf{J}(0)\mathbf{U}$. Computing the lengths of these axes directly is numerically inadequate, because directions that are stretched and directions that are compressed produce very large and very small numbers. Instead, the growth of each axis is computed

at every iteration, and the axes are orthonormalized again before the next one, in the following steps [1]:

1) Choose an orthonormal basis $\{\mathbf{w}_1(0), \ldots, \mathbf{w}_K(0)\}$ and compute $\mathbf{z}_k(0) = \mathbf{J}(0)\mathbf{w}_k(0)$, $1 \leq k \leq K$. These vectors are axes of the new ellipsoid, but they are not necessarily orthogonal.
2) Orthogonalize them by the Gram–Schmidt process,

$$\mathbf{y}_k(1) = \mathbf{z}_k(0) - \sum_{i=1}^{k-1} \frac{\mathbf{z}_k(0) \cdot \mathbf{y}_i(1)}{\|\mathbf{y}_i(1)\|^2}\, \mathbf{y}_i(1), \qquad (8)$$

where "$\cdot$" denotes the scalar product.
3) Normalize the vectors, $\mathbf{w}_k(1) = \mathbf{y}_k(1)/\|\mathbf{y}_k(1)\|$, which gives the orthonormal basis for the next iteration.
4) Apply the Jacobian at the next point of the orbit, $\mathbf{z}_k(1) = \mathbf{J}(1)\mathbf{w}_k(1)$, and repeat steps 2 and 3. After n repetitions, the vectors $\mathbf{y}_k(1), \ldots, \mathbf{y}_k(n)$ are available.

The Lyapunov exponent in the k-th direction is then

$$\lambda_k = \lim_{n\to\infty} \frac{\ln \|\mathbf{y}_k(1)\| + \cdots + \ln \|\mathbf{y}_k(n)\|}{n}. \qquad (9)$$

Only the largest exponent, λ_1, is used in this work. For $\alpha = 1.4$ and $\beta = 0.3$, the map (1) has $\lambda_1 \approx 0.42$ [22].

Based on these concepts, each orbit is classified as

$$\begin{cases} \text{divergent}, & \text{if the orbit is unbounded,} \\ \text{chaotic}, & \text{if it is bounded and } \lambda_1 > 0, \\ \text{periodic}, & \text{if it is bounded and } \lambda_1 \leq 0, \end{cases} \qquad (10)$$

where the periodic class also includes convergence to a fixed point, because the criterion distinguishes only the presence or absence of chaos and not the period of the limit orbit. The numerical detection of unbounded orbits is described in Section III-C.

D. FIR Design Windows

This subsection describes how the coefficients c_j are obtained and which design windows are compared. The coefficients are obtained by the window method [23], [25]: the impulse response of the ideal filter, which is infinitely long, is conveniently delayed and truncated by a window of N_s samples. The shape of the window sets a trade-off between the stopband attenuation and the width of the transition band of the resulting filter: windows with lower side lobes give higher attenuation at the cost of a wider transition band [25].

Seven windows are considered (Table I). The values listed are the peak side-lobe levels of the discrete-time Fourier transform of each window, relative to the main-lobe peak [25]; in the window method, lower side lobes give higher stopband attenuation of the designed filter [23]. The values range from -13 dB for the rectangular window, whose abrupt truncation produces strong side lobes, to -92 dB for the four-term Blackman–Harris window, whose cosine-sum coefficients are chosen to cancel side lobes [25].

The seven windows cover a broad range of this trade-off: the rectangular window as the worst case; the Bartlett and Hann windows as classical intermediate cases; the Hamming window, adopted as a reference because it was used in the previous works on the filtered Hénon map [11], [17]; and the Blackman and Blackman–Harris windows as high-attenuation cases. They were chosen under the hypothesis that greater spectral attenuation favors the preservation of chaos by reducing side-lobe leakage into the rejected band. The Kaiser window [20] has a shape parameter $\beta_K \geq 0$ that varies the trade-off continuously without changing the filter order: for $\beta_K = 0$ it is the rectangular window, and for $\beta_K \in [2, 14]$ its peak side-lobe level, computed here for a window of 101 samples, goes from about -19 dB to -106 dB, a range that contains the side-lobe levels of the Bartlett, Hann, Hamming, Blackman, and Blackman–Harris windows in Table I. The sweep of β_K therefore provides a continuous path between the fixed windows.

TABLE I
FIR WINDOWS EVALUATED AND THEIR PEAK SIDE-LOBE LEVELS

Window	Side lobe (dB)	Characteristic
Rectangular (boxcar)	-13	Minimum reference
Bartlett	-27	Triangular
Hann	-32	Raised cosine
Hamming	-43	Standard reference
Blackman	-58	High attenuation
Blackman–Harris	-92	Maximum attenuation
Kaiser (β_K variable)	variable	Parametric

Source: adapted from [25].

E. Synchronization Versus Chaos: Motivation

This subsection explains why a filter in the feedback loop is needed in coherent CBCS and why its presence makes the verification of chaos necessary. It is a motivation for the study: the sweeps of Sections III and IV do not transmit any message.

In a coherent CBCS of the master–slave type [5], let $\mathbf{x}(n)$ and $\mathbf{y}(n)$ denote the states of the transmitter (master) and of the receiver (slave), and let $\mathbf{e}(n) \triangleq \mathbf{y}(n) - \mathbf{x}(n)$ be the synchronization error. When the nonlinearity acts on a single scalar signal extracted from the state, as $x_1(n)$ in (1), the error obeys the linear recursion

$$\mathbf{e}(n+1) = \mathbf{A}\,\mathbf{e}(n), \qquad (11)$$

where $\mathbf{A}$ is the synchronization matrix. Synchronization is attained when every eigenvalue γ_k of $\mathbf{A}$ lies strictly inside the unit circle, $|\gamma_k| < 1$ [15]. Over an ideal, distortion-free channel, this condition is enough for the receiver to track the transmitter and for the message to be recovered after an initial transient.

Real channels, however, have a finite passband and act as filters that reshape the spectrum of the transmitted signal [9]. Because chaotic carriers spread their energy continuously over a wide band [2], even a moderate truncation of the spectrum makes the slave evolve under a different effective nonlinearity, so the synchronization error no longer vanishes and the recovered message no longer follows the original one [15]. Confining the transmitted spectrum to the channel passband before transmission is therefore a precondition for coherent detection over band-limited channels [10], [11].

Figs. 2 and 3 illustrate this argument. In both, the message encodes the ASCII text "hello" (8 bits per character, $+1$ for

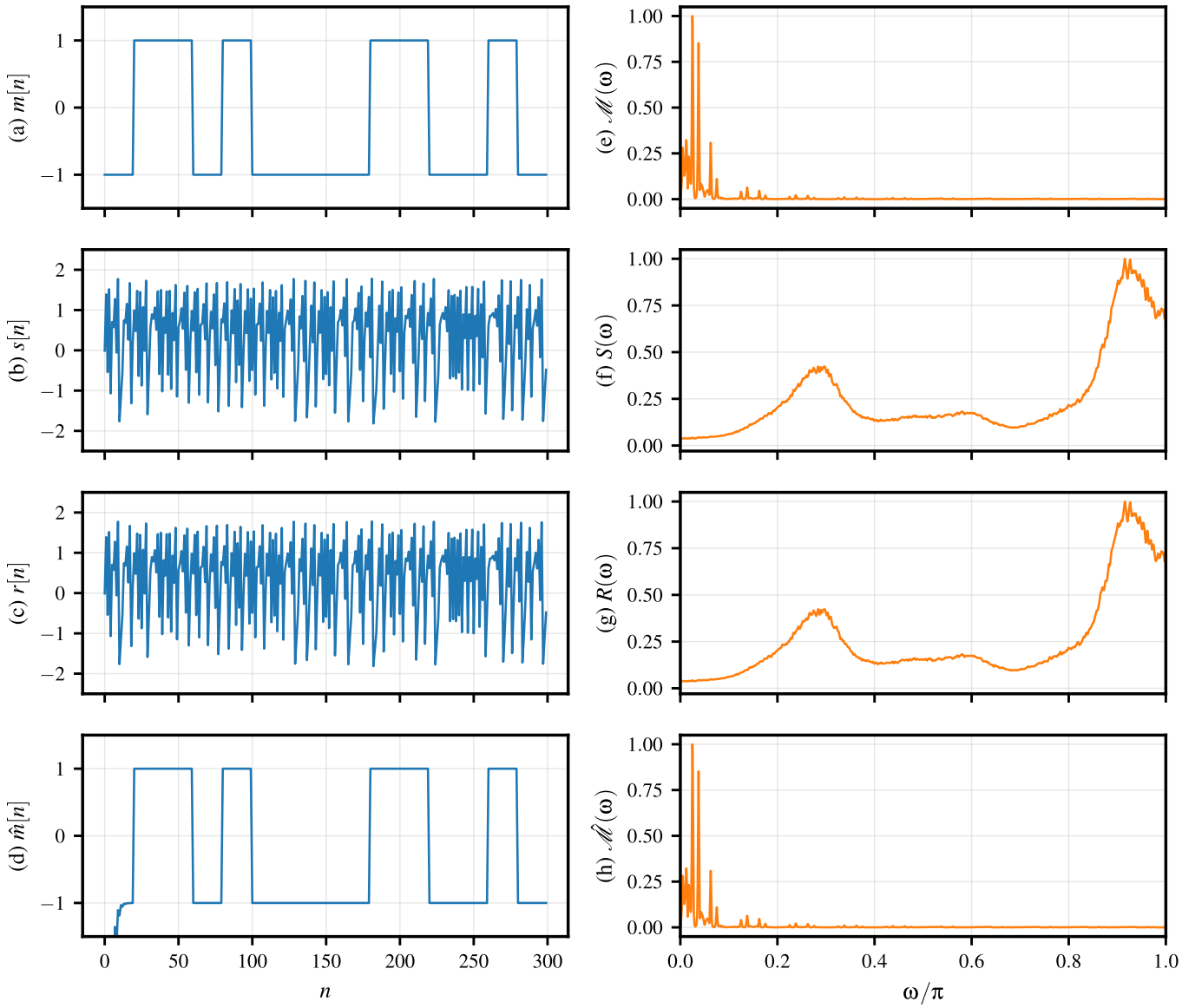


Fig. 2. Illustration used as motivation: ideal-channel communication over $N = 10^6$ samples with the generator (1), a ± 1 non-return-to-zero (NRZ) binary phase-shift keying (BPSK) message (ASCII text "hello", bit period of 20 samples), and transmitted signal $s[n] = x_1[n] + \mu\, m[n]$ with modulation depth $\mu = 0.01$: message $m[n]$, transmitted $s[n]$, received $r[n]$, and recovered $\hat{m}[n]$ (left, time domain) with their normalized Welch power spectral densities (PSDs) (right). The channel is ideal (identity, zero delay); the initial transient of $\hat{m}[n]$ is the synchronization of the slave.

bit 1 and -1 for bit 0), each bit is held for 20 samples, and the block is repeated to fill $N = 10^6$ samples. The mean squared error between $m[n]$ and $\hat{m}[n]$ excludes the first 200 samples, which contain the synchronization transient, and the bit error rate (BER) decides each bit by the sign of the mean of $\hat{m}[n]$ over its 20 samples. Over the ideal channel (Fig. 2), the received signal is a copy of the transmitted one, the recovered spectrum $\hat{\mathcal{M}}(\omega)$ follows the message spectrum $\mathcal{M}(\omega)$, and the mean squared error is 8.7×10^{-31}, that is, machine precision.

In Fig. 3, the channel is a 201-tap linear-phase lowpass FIR filter designed with the Hamming window and cutoff $0.99\,\pi$, whose group delay of 100 samples is compensated in the metrics. Even this channel, which removes only the highest frequencies, prevents synchronization when the generator is the map (1) without a filter in the loop (Fig. 3(a)): the BER is 0.4324. When the generator has an FIR filter in the loop (9 coefficients, $N_z = 8$, $\omega_c = 0.5\,\pi$, Hamming window) and the transmitted carrier is the filtered variable x_3, the same channel gives a BER of 0 and a mean squared error of 2.0×10^{-5} (Fig. 3(b)). These numbers refer only to this configuration and are reported as motivation.

Inserting the filter in the feedback loop, rather than only at the channel interface, also changes the synchronization matrix $\mathbf{A}$, as stated next.

Lemma 1 (Preservation of synchronization under FIR feedback filtering, adapted from [15])**.** *Let the unfiltered master–slave pair satisfy the synchronization condition $|\gamma_k| < 1$ for every eigenvalue γ_k of $\mathbf{A}$. Inserting the same FIR filter in the feedback loop of the master and of the slave, as in* (4)*, only adds null eigenvalues to the synchronization matrix. Hence*

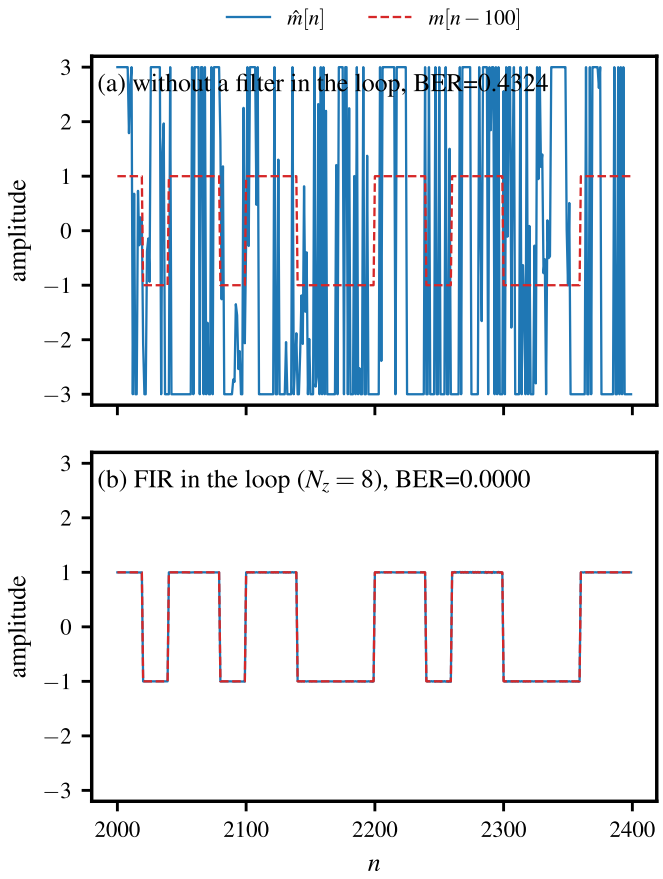


Fig. 3. Illustration used as motivation: recovered message $\hat{m}[n]$ (clipped to ± 3) and delayed message $m[n-100]$ over a lowpass FIR channel (201 taps, Hamming window, cutoff $0.99\,\pi$). (a) Generator (1) without a filter in the loop, BER $= 0.4324$. (b) Generator with an FIR filter in the loop ($N_z = 8$, $\omega_c = 0.5\,\pi$, Hamming window), BER $= 0$.

the original condition remains valid, independently of the coefficients c_j [11], [15].

Proof sketch. The nonlinearity of master and slave is driven by the same transmitted signal, so it cancels in the error, and the filter states (the filter output and its delay line) do not feed back into the error of the original states. Ordering the error vector as [original states; filter states], the synchronization matrix is block lower triangular, with diagonal blocks $\mathbf{A}$ and a shift matrix of zero diagonal. Its eigenvalues are therefore those of $\mathbf{A}$ together with zeros [15] (Theorem 3.1; the block-triangular argument is detailed for infinite impulse response (IIR) filters in its Theorems 6.1 and 6.2).

By Lemma 1, the filter preserves the synchronization of the master–slave pair for any choice of coefficients, provided that the unfiltered pair synchronizes. The lemma concerns only the linear stability of the error recursion (11), however, and says nothing about whether the master's own orbit remains chaotic once the same coefficients enter the nonlinear dynamics (4) and its Jacobian (7). Synchronization and chaotic behavior are thus independent requirements: the former is guaranteed by construction, whereas the latter is not, as shown numerically in [14], [16], [17]. A frequency response $H(e^{j\omega})$ that fits the channel passband does not ensure, by itself, that the filtered generator remains chaotic. The dynamics of each design must be verified, which is the purpose of the methodology described next.

III. Methodology

The research adopts a quantitative-experimental approach: for each filter design, the expanded system (4) is simulated, λ_1 is estimated, and each grid point is classified by (10). All the tools were implemented in the open-source Python package `chaotic_pfc` (BSD-3-Clause license) [24].

A. Filter Types and Parameter Grid

Four filter types are designed with the seven windows of Table I by the routine `firwin` of the SciPy library [26]. For lowpass and highpass filters, ω_c is the cutoff frequency; for bandpass and bandstop filters, it is the center of a band of fixed total width $\Delta\omega = 0.05\,\pi$, placed symmetrically around ω_c. On the normalized axis $\omega/\pi \in (0, 1)$, this width is 5% of the range and spans about five steps of the cutoff grid, so the band is narrow enough to keep the filter frequency-selective and wide enough to be resolved by the sweep at every ω_c. Fixing $\Delta\omega$ also isolates the effect of ω_c; the width is left as a variable for future work (Section V).

Because it is the frequency response, and not the individual coefficients, that must be matched to the channel (Section II-E), the sweep variables are the filter order N_z and the cutoff frequency ω_c. For lowpass and bandpass filters, $N_z \in \{1, 2, \ldots, 40\}$ and 100 values of ω_c/π linearly spaced in the open interval $(0, 1)$ are swept, giving $4{,}000$ grid points per window–filter configuration. For highpass and bandstop filters, the parity restriction of the window method requires even N_z [23], because a filter with an odd order has a forced zero at $\omega = \pi$, which a passband that includes $\omega = \pi$ does not allow; thus $N_z \in \{2, 4, \ldots, 40\}$, with $2{,}000$ points per configuration. The six fixed windows give $4 \times 6 = 24$ configurations, and the Kaiser window is swept over 25 values of $\beta_K \in [2.0, 14.0]$, with step 0.5, for the four filter types, adding 100 configurations, for a total of 124 sweeps and $372{,}000$ grid points. For the Hann and Bartlett windows with lowpass and bandpass filters, `firwin` returns non-finite coefficients for part of the grid; these points are classified as divergent because of the filter design, not of the map dynamics.

B. Lyapunov Estimator and Initial Conditions

Computing all K exponents by the method of Section II-C costs $\mathcal{O}(K^3)$ operations per iteration, dominated by the Gram–Schmidt orthogonalization. Since only λ_1 is needed, the sweeps propagate a single tangent vector, multiplied by $\mathbf{J}(n)$ at each iteration and renormalized once every ten iterations, which reduces the cost to $\mathcal{O}(K)$ per iteration. The formulation follows Wolf *et al.* [27], and the use of a renormalization period follows Benettin *et al.* [28], [29]. Algorithm 1 summarizes the procedure.

As in [14], λ_1 is taken as the largest of the K finite-time exponents. To reduce the cost, a single tangent vector is

Algorithm 1 Single-vector estimation of λ_1 with periodic renormalization.

Input: initial condition $\mathbf{x}_0$; burn-in iterations N_{burn}; accumulation cap N_{map}; tolerance ε.
$\mathbf{w} \leftarrow [1\ 0\ \cdots\ 0]^T$; $s \leftarrow 0$; $\mathbf{x} \leftarrow \mathbf{x}_0$.
for $i = 1$ **to** N_{burn} **do**
  $\mathbf{x} \leftarrow \mathbf{F}(\mathbf{x})$ (discard transient)
  if $\mathbf{x}$ is non-finite **then**
    return DIVERGENT (grid point)
  end if
end for
for $i = 1$ **to** N_{map} **do**
  $\mathbf{w} \leftarrow \mathbf{J}(\mathbf{x})\,\mathbf{w}$ (tangent-vector evolution at the current state)
  $\mathbf{x} \leftarrow \mathbf{F}(\mathbf{x})$ (orbit evolution)
  if $\mathbf{x}$ is non-finite **then**
    return NaN (divergent orbit)
  end if
  if i is a multiple of 10 **then**
    $r \leftarrow \|\mathbf{w}\|$; $s \leftarrow s + \ln\max(r, 10^{-300})$; if $r > 10^{-300}$ then $\mathbf{w} \leftarrow \mathbf{w}/r$
    every 100 iterations with $i \geq 500$: stop if s/i varied less than ε in two consecutive checks
  end if
end for
return $\lambda_1 = s/i$

propagated first, and grid points whose mean estimate satisfies $|\lambda_1| < 5 \times 10^{-3}$ are recomputed with the full spectrum of Section II-C, using the modified Gram–Schmidt algorithm; 34,147 of the 372,000 grid points (9.18%) were recomputed.

Accumulation stops early when the estimate of λ_1 varies less than the tolerance $\varepsilon = 10^{-3}$ in two consecutive checks, made every 100 iterations after the 500th. Requiring two checks avoids premature stops in windows of chaotic intermittency, in which λ_1 oscillates for hundreds of iterations before stabilizing.

For each grid point (N_z, ω_c), $N_{\text{IC}} = 25$ initial conditions are drawn near the positive fixed point $\mathbf{p}^+$ of the expanded system (Section II-B): each component is $p_k^+ + 0.1\,U$, with U uniformly distributed in $[0, 1)$. For each initial condition, $N_{\text{burn}} = 500$ iterations are discarded as transient, and λ_1 is accumulated for at most $N_{\text{map}} = 3{,}000$ iterations. The value assigned to the grid point is the mean of λ_1 over the initial conditions.

The value $N_{\text{burn}} = 500$ is much longer than the transients of the unfiltered map and accommodates longer transients near bifurcation boundaries. The value N_{map} acts as a cap: with early stopping, the median number of accumulated iterations per configuration lies between about 700 and 1,200, with a floor of 700, and the cap is reached in at most 0.06% of the points, in only two of the 124 configurations. In all experiments, $\alpha = 1.4$, $\beta = 0.3$, and the seed of the pseudo-random generator is 42.

C. Classification and Chaotic-Coverage Metrics

Each grid point is classified by (10). Numerically, an orbit is taken as unbounded when its state becomes non-finite (NaN or infinity); from that iteration on, every term of (9) is non-finite, so the simulation can be stopped. The divergence rule has three steps. First, the estimator returns a non-finite value for an initial condition as soon as its state becomes non-finite. Second, a grid point is divergent when at least one of its 25 initial conditions becomes non-finite during the burn-in. Third, an initial condition that becomes non-finite only after the burn-in is discarded, and the value of the grid point is the mean of λ_1 over the remaining initial conditions. This step differs from [14], in which a grid point is divergent when any of its initial conditions becomes non-finite. Points with non-finite filter coefficients (Section III-A) are also divergent.

Let N_{grid} be the number of grid points of a configuration, N_b the number of bounded (chaotic or periodic) points, and N_c the number of chaotic points. The *conditional* fraction $\rho_c^{\text{fin}} = N_c/N_b$ is the share of chaotic points among the bounded ones; it measures the tendency to chaos once divergence is excluded and is relevant when N_z and ω_c can be adjusted to reach a non-divergent operating point. The *absolute* fraction $\rho_c^{\text{abs}} = N_c/N_{\text{grid}}$ is the share of the whole design space that yields a chaotic generator and is relevant when a broad operating range is sought. Using fractions also makes the comparison between filter types independent of their different grid sizes (Section III-A).

D. Statistical Treatment

Two levels are kept separate. At the *point* level, the value of λ_1 of each grid point is the mean over its N_{IC} initial conditions, and the dispersion over the ensemble is stored as metadata. At the *regional* level, the chaotic region of each configuration is summarized by the mean $\bar{\lambda}_1$ and the standard deviation σ_{λ_1} of λ_1 over its N_c chaotic points; thus σ_{λ_1} measures how homogeneous λ_1 is across the chaotic region, not the estimation noise at one point. A 95% confidence interval for $\bar{\lambda}_1$ is obtained by nonparametric bootstrap [30] with 1,000 resamples and seed 42. These regional statistics are the ones reported in Tables II and III. Because the chaotic points are selected by $\lambda_1 > 0$, this interval is positive by construction, and because neighboring grid points are correlated, the i.i.d. bootstrap underestimates the uncertainty; the interval is therefore used only to describe dispersion. The robustness of the classification is assessed by three other checks: the fraction of chaotic points whose ensemble mean of λ_1 exceeds two standard errors, $2s/\sqrt{N_{\text{IC}}}$, where s is the standard deviation of λ_1 over the initial conditions of the point; the sensitivity of N_c to the classification threshold (0, 10^{-3}, and 5×10^{-3}); and the repetition of the configurations of Table II with ten seeds of the initial-condition generator (1 to 9 and 42).

E. Computational Architecture and Reproducibility

Reliable estimates of λ_1 require long simulations and an ensemble of initial conditions [27], and each sweep has thousands of grid points: a 40×100 grid with 25 initial conditions

requires 10^5 runs of the estimator. For the full-spectrum recomputation, the cost per grid point is $\mathcal{O}(N_{\mathrm{IC}} \cdot N_{\mathrm{map}} \cdot K^3)$, which for $N_z = 40$ ($K = 41$) approaches 5×10^9 floating-point operations. Three decisions make the study tractable. First, the numerical cores are compiled just in time by Numba [31], which translates Python functions into machine code through LLVM and reduced the execution time by one to two orders of magnitude relative to pure Python in the package benchmarks [24]. Second, the grid points are processed in parallel on all processor cores with `prange` loops. Third, each sweep is stored as a compressed NumPy (`.npz`) file, so that figures and tables can be regenerated without repeating the sweeps.

Each `.npz` file contains the ensemble mean of λ_1 per grid point and its standard deviation, the filter orders and coefficients, the cutoff grid, the window and filter-type labels, the number of iterations actually used, and the simulation parameters (N_{burn}, N_{map}, number of initial conditions, α, β, seed, β_K, early-stop settings, and number of recomputed points). The files are versioned in `data/sweeps/`; the figures are generated by `src/chaotic_pfc/analysis/sweep_plotting.py` and the statistics by `chaotic_pfc/analysis/stats.py`. Automated tests in `tests/`, run by continuous integration at every change of the code, verify the identity $\lambda_1 + \lambda_2 = \ln|\beta|$ for (1), the value of λ_1 against published references [22], [27], and the designed filters (coefficient sum and zero placement). The package has a Git version tag, a `CITATION.cff` file following the FORCE11 recommendations for software citation, and Sphinx documentation published on Read the Docs [24].

IV. Results and Discussion

This section presents the results of the 124 sweeps.

A. Validation of the Estimator

The estimator was first validated on the map (1) without filter ($N_s = 1$, $c_0 = 1$), with $\alpha = 1.4$ and $\beta = 0.3$, for which $\lambda_1 \approx 0.42$ [22], [27]. With the ensemble of Section III-B and early stopping disabled, it gave $\bar{\lambda}_1 = 0.4182$, with a relative error below 10^{-2}, which is enough to separate the classes of (10). The identity $\lambda_1 + \lambda_2 = \ln|\beta|$ (Section II-C) was satisfied to machine precision in all unfiltered configurations tested, which confirms the correctness of the estimator independently of the reference value. This validation used the full-spectrum estimator; the sweeps use the single-vector path of Algorithm 1, with early stopping.

B. Configurations With Largest Chaotic Coverage

Table II shows, for each filter type, the configuration that maximizes N_c, considering both the fixed windows and the Kaiser window with the β_K that maximizes N_c. The second configuration in decreasing order of N_c is also shown: a fixed window for lowpass, highpass, and bandpass filters, and the Kaiser window for the bandstop filter, whose first position is the rectangular window.

The Kaiser window has the largest N_c in three of the four filter types, in both ρ_c^{fin} and ρ_c^{abs}. Repeating the configurations of Table II with ten seeds gives, as mean $\pm$ standard deviation of N_c, 975.3 ± 2.7 (Kaiser) and 957.8 ± 3.9 (Blackman–Harris) for lowpass, 147.6 ± 1.2 and 142.9 ± 2.1 for highpass, and 435.8 ± 3.5 and 418.6 ± 3.8 for bandpass; the Kaiser configuration has the larger N_c for every seed. For bandstop, the values are 116.8 ± 2.5 (rectangular) and 57.0 ± 2.2 (Kaiser). The ordering in Table II is therefore not a seed effect. The comparison must still be read with caution, because each fixed window is a single configuration, whereas the Kaiser window was evaluated at 25 values of β_K and the best one was selected (Section IV-E). Among the fixed windows, Blackman–Harris is the best for lowpass, highpass, and bandpass filters, which is consistent with its lowest side lobes (-92 dB). Over the six fixed windows ($n = 6$), the Spearman correlation coefficient r_s between the side-lobe attenuation of Table I and N_c, with its exact two-sided permutation p-value over the 720 permutations, is $r_s = +0.83$ ($p = 0.0583$) for lowpass, $+0.77$ ($p = 0.1028$) for highpass, $+1.00$ ($p = 0.0028$) for bandpass, and -0.26 ($p = 0.6583$) for bandstop. The correlation is computed on N_c, not on ρ_c^{fin}, and no correction is applied for the four tests with $n = 6$. It is significant at the 5% level only for the bandpass filter, which remains significant with a Bonferroni correction ($0.05/4 = 0.0125$). For lowpass filters designed with the Hamming and Blackman windows, Borges and Eisencraft [17] observed similar curves of λ_1 versus cutoff frequency and concluded that the window seemed to have little influence on the generation of chaos. In the present sweep, in contrast, the Blackman window gives 801 chaotic points and the Hamming window 450 (Table III). The two studies differ in the grid of orders and cutoff frequencies, in the ensemble of initial conditions (10 drawn in $[0, 1]$ in [17], 25 drawn near $\mathbf{p}^+$ here), and in the comparison criterion (curves of λ_1 there, counts of chaotic points here). The lowpass filter with the Hamming window also corresponds to Experiment V of [14], which swept the same range of N_z.

The bandstop filter behaves differently: with the rectangular window it reaches $\rho_c^{\mathrm{fin}} = 65.2\%$, the highest conditional fraction among the four filter types for this window (19.5% lowpass, 3.7% bandpass, 1.3% highpass), but its absolute coverage is only $\rho_c^{\mathrm{abs}} = 5.9\%$, because a large share of its grid is divergent at higher orders (Section IV-C). Since this filter rejects only a narrow band ($\Delta\omega = 0.05\,\pi$) away from $\omega = 0$, its DC gain G remains close to one. The lowpass filter with the Kaiser window ($\beta_K = 13.5$) reaches the highest absolute coverage among the 124 configurations, $\rho_c^{\mathrm{abs}} = 24.5\%$, about 4.2 times that of the rectangular bandstop filter, or 8.3 times in number of chaotic points (979 against 118).

C. Classification Maps

Figs. 4–7 show the classification maps in the $(N_z, \omega_c/\pi)$ plane for the configuration of largest N_c of each filter type, with chaotic points in red, periodic points in blue, and divergent points in gray. In the lowpass map (Fig. 4), chaos lies below a boundary in ω_c/π that decreases with N_z, and the orbits diverge above it. For N_z from 3 to 7, chaotic points reach $\omega_c/\pi \approx 1$ (up to 0.78 for $N_z = 4$), whereas the largest chaotic cutoff is 0.44 for $N_z = 10$, 0.14 for $N_z = 20$, and

TABLE II
CONFIGURATIONS OF LARGEST N_c PER FILTER TYPE, WITH MEAN, STANDARD DEVIATION, AND CONFIDENCE INTERVAL OF λ_1 OVER THE CHAOTIC POINTS

Filter	**Window**	N_c	ρ_c^{fin}	ρ_c^{abs}	$\bar{\lambda}_1$	σ_{λ_1}	95% CI of $\bar{\lambda}_1$
Lowpass	Kaiser (β_K=13.5)	979	77.6%	24.5%	0.0239	0.0231	[0.0223, 0.0254]
	Blackman–Harris	961	74.8%	24.0%	0.0211	0.0232	[0.0197, 0.0227]
Highpass	Kaiser (β_K=11.5)	149	8.1%	7.5%	0.0178	0.0263	[0.0138, 0.0219]
	Blackman–Harris	144	7.9%	7.2%	0.0190	0.0269	[0.0151, 0.0237]
Bandpass	Kaiser (β_K=13.5)	434	13.0%	10.8%	0.0286	0.0297	[0.0258, 0.0315]
	Blackman–Harris	418	12.3%	10.4%	0.0261	0.0301	[0.0235, 0.0293]
Bandstop	Rectangular	118	65.2%	5.9%	0.0709	0.0491	[0.0627, 0.0800]
	Kaiser (β_K=4.0)	58	39.2%	2.9%	0.0726	0.0336	[0.0626, 0.0798]

95% confidence intervals obtained by bootstrap [30]. Full per-configuration tables are regenerated from the versioned sweep files by the released package [24].

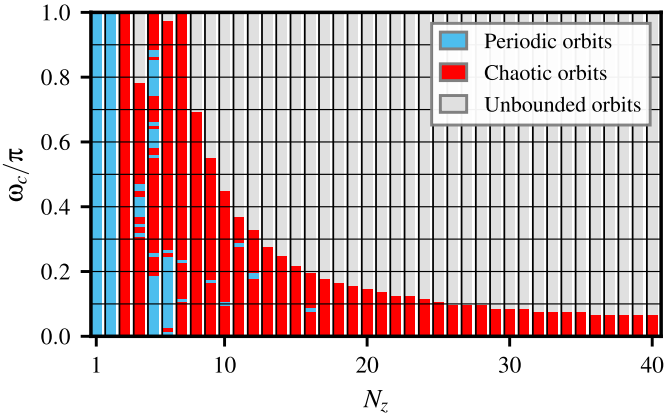


Fig. 4. Classification map for the lowpass filter, Kaiser window ($\beta_K = 13.5$). Chaos lies below a cutoff boundary that decreases with N_z, with divergence above it; periodic islands (shrimps) appear at low orders.

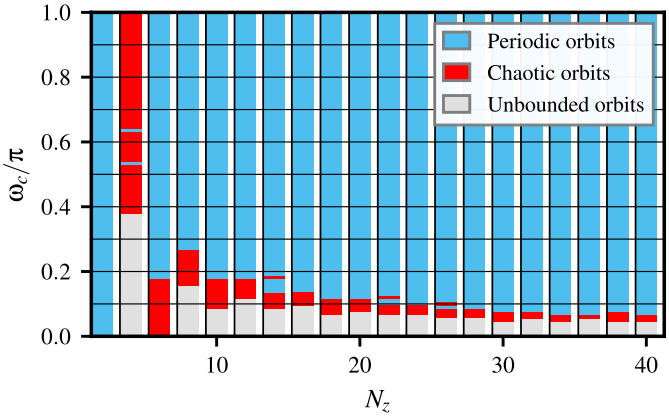


Fig. 5. Classification map for the highpass filter, Kaiser window ($\beta_K = 11.5$). Chaos concentrates at $\omega_c/\pi \leq 0.26$ for $N_z \geq 6$ and spans the whole cutoff range only for $N_z = 4$ ($\rho_c^{\text{abs}} = 7.5\%$).

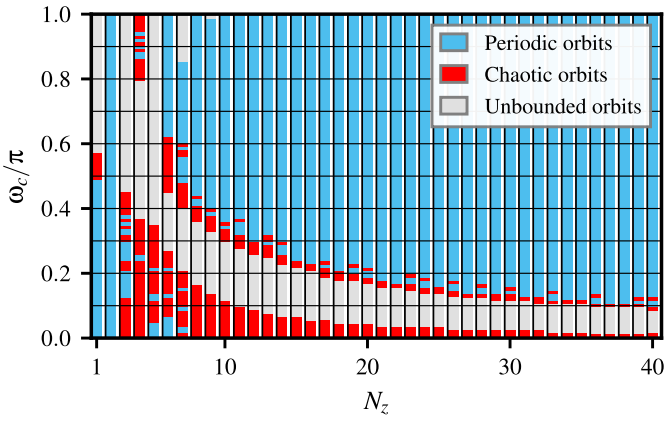


Fig. 6. Classification map for the bandpass filter, Kaiser window ($\beta_K = 13.5$), $\Delta\omega = 0.05\,\pi$. Fragmented chaotic region, more sensitive to ω_c; maximum λ_1 equal to 0.196.

0.06 for $N_z = 40$; for $N_z = 1$ and 2 no point is chaotic. In the highpass map (Fig. 5), chaos concentrates at $\omega_c/\pi \leq 0.26$ for $N_z \geq 6$ and spans the whole cutoff range only for $N_z = 4$ ($0.38 \leq \omega_c/\pi \leq 1$); the largest λ_1, 0.106, occurs at $N_z = 6$ and $\omega_c/\pi = 0.01$, the lower edge of the swept interval. The lowpass and highpass filters therefore differ mainly in how the chaotic region depends on N_z: at high orders, both have chaos only at low cutoff frequencies. The bandpass map (Fig. 6) shows a fragmented chaotic region, more sensitive to ω_c, which is consistent with the narrow band imposed by the filter. In the bandstop map (Fig. 7), chaos appears in isolated bands and divergence dominates the grid: only $N_z = 2$ has no divergent points, and for $N_z = 4$, 8, 12, and 16 the whole row diverges.

These maps are consistent with two observations of [14], [16]. First, the DC gain G sets the positive fixed point (6), so filter types with different G place the orbits in different regions of the state space [14]. Second, the thin boundaries between chaotic and periodic regions and the periodic islands inside the chaotic regions resemble the period-doubling cascades and the shrimps [18] documented for the filtered Hénon map by Borges and Eisencraft [16]; such structures appear in the four filter types studied here.

The filter types also differ in how the grid is divided among the three classes. Pooling the 124 sweeps, the numbers of chaotic, periodic, and divergent points are: lowpass, 20,906, 17,701, and 85,393 out of 124,000 (68.87% divergent); highpass, 2,900, 54,902, and 4,198 out of 62,000 (6.77%); bandpass, 10,219, 100,990, and 12,791 out of 124,000 (10.32%); and bandstop, 1,598, 3,035, and 57,367 out of 62,000 (92.53%). Chaos is therefore limited mainly by divergence in

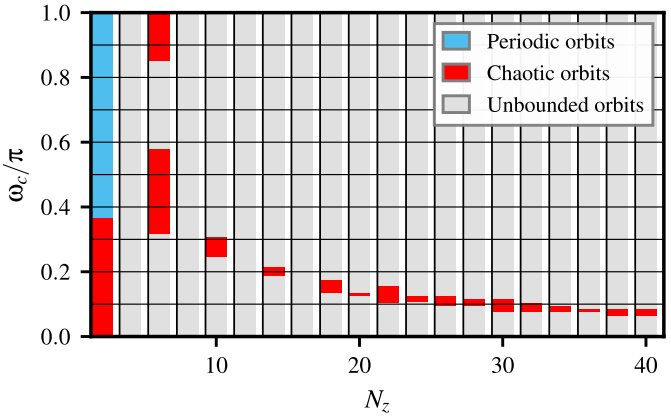


Fig. 7. Classification map for the bandstop filter, rectangular window, $\Delta\omega = 0.05\,\pi$ ($\rho_c^{\text{fin}} = 65.2\%$, $\rho_c^{\text{abs}} = 5.9\%$). Only $N_z = 2$ has no divergent points.

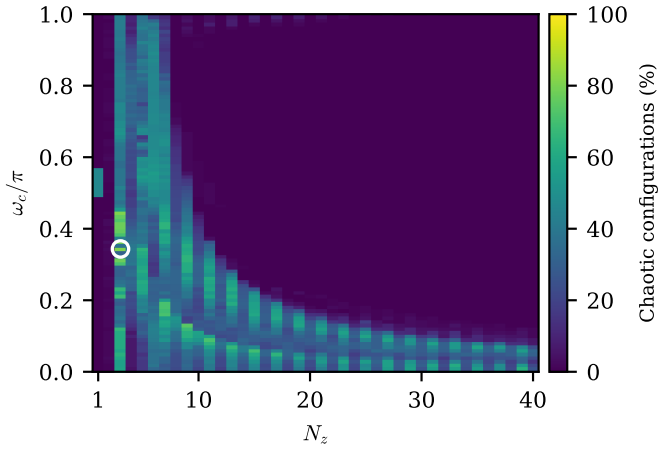


Fig. 8. Aggregate chaotic density over the 124 sweeps: percentage of the configurations containing each grid point in which it is chaotic (124 configurations for even N_z, 62 for odd N_z, which have only lowpass and bandpass filters). The marker (○) indicates the maximum, 80.6% at $N_z = 3$ and $\omega_c/\pi \approx 0.34$.

lowpass and bandstop filters and mainly by periodic orbits in highpass and bandpass filters. Aggregated over all its configurations, the bandstop filter has $\rho_c^{\text{fin}} = 34.5\%$ (1,598 of 4,633 bounded points), below the lowpass filter (54.2%) and above the bandpass (9.2%) and highpass (5.0%) filters; the value of 65.2% is specific to the rectangular window. The divergent fraction of 92.53% of the bandstop filter is a result of this study and does not constitute a recommendation for the use of this filter type.

D. Aggregate Chaotic Density

To identify the regions of the $(N_z, \omega_c/\pi)$ plane that are chaotic regardless of the design, the chaotic points of the 124 sweeps were aggregated. Because highpass and bandstop filters have only even N_z, a grid point with even N_z belongs to 124 configurations and one with odd N_z to 62 (lowpass and bandpass only). Fig. 8 therefore shows, at each grid point, the percentage of the configurations containing it in which it is chaotic.

For $N_z \geq 8$, chaos is most frequent at low cutoff frequencies: on average, a grid point is chaotic in 31.3% of the configurations for $\omega_c/\pi \leq 0.1$, in 16.8% for $0.1 < \omega_c/\pi \leq 0.2$, in 5.9% for $0.2 < \omega_c/\pi \leq 0.4$, and in less than 1.5% above $\omega_c/\pi = 0.4$. For $N_z \leq 7$, the fraction varies less with the cutoff, from 19.8% to 29.4%. The maximum, 80.6%, occurs at $N_z = 3$ and $\omega_c/\pi \approx 0.34$ (50 of 62 configurations); among even orders, it occurs at $N_z = 10$ and $\omega_c/\pi \approx 0.11$ (71 of 124, 57.3%). Odd and even orders are not directly comparable, because the odd ones include only the lowpass and bandpass filters, which have more chaotic points; this explains the alternation between neighboring columns in Fig. 8.

The choice of window matters mainly at low and moderate orders. The six fixed windows give the same class (chaotic, periodic, or divergent) at 25% to 29% of the lowpass grid points for $N_z \leq 10$ and at 88% to 91% for $N_z > 20$; the corresponding values are 39%–58% and 86%–90% for bandpass, 57%–83% and 93%–94% for highpass, and 78%–82% and 97%–98% for bandstop (points with non-finite coefficients count as divergent). This convergence is expected, because the designed filters approach the ideal one as N_z grows [23].

Because the regime boundaries are finely interleaved, with period-doubling cascades alternating with shrimps at progressively smaller scales [16], [18], a small change of N_z or ω_c can move a design across a boundary. We therefore suggest, without having measured it here, that the distance from an operating point to the nearest boundary be considered in the design.

E. Effect of the Kaiser Parameter β_K

The values of β_K that maximize N_c (Table II) are 13.5 for lowpass and bandpass filters, 11.5 for highpass, and 4.0 for bandstop. For the first three types, the maximum lies at high β_K, where the side lobes of the window are strongly attenuated [23], which is consistent with the advantage of the Blackman–Harris window among the fixed windows (Section IV-B). Fig. 9 shows ρ_c^{abs} versus β_K. For the lowpass filter, ρ_c^{abs} grows from 7.1% at $\beta_K = 2$ to 23.0% at $\beta_K = 10$ and stays between 22.9% and 24.5% up to $\beta_K = 14$. For the bandpass filter, it grows from 4.0% to values between 9.9% and 10.9% for $\beta_K \geq 10$, and for the highpass filter, from 2.0% to a maximum of 7.5% at $\beta_K = 11.5$, followed by values between 6.4% and 7.0%. Differences between neighboring values on these plateaus, such as 973 and 979 chaotic points for the lowpass filter at $\beta_K = 13.0$ and 13.5, are of the order of the seed variability (Section IV-B); the β_K of largest N_c in Table II is therefore one value inside a plateau rather than a sharp optimum.

For the bandstop filter, the largest N_c occurs with the rectangular window and, among the Kaiser configurations, at $\beta_K = 4.0$, but its curve in Fig. 9 is nearly flat, between 2.0% and 2.9%. One hypothesis, not tested here, is that, when the filter removes only a narrow band of the spectrum, preserving most of the spectral energy in the feedback loop matters more than the depth of the stopband attenuation, which would agree with the result of the rectangular window, of lowest attenuation, for this filter type. The effect of β_K is therefore

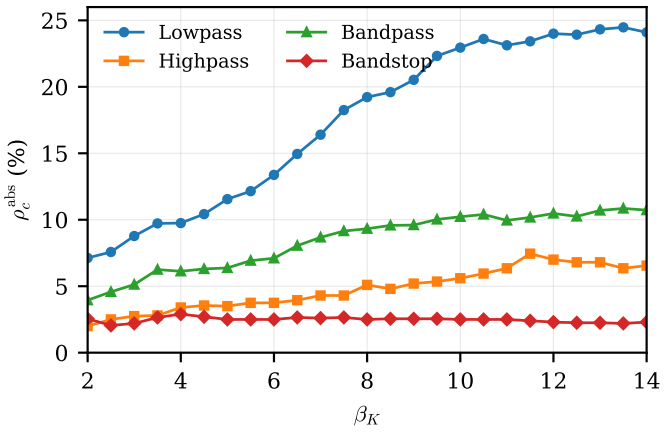


Fig. 9. Absolute chaotic coverage ρ_c^{abs} versus the Kaiser shape parameter β_K for the four filter types.

not the same for the four filter types; it depends on the spectral mask imposed.

F. Statistical Confidence and Ranking

The 95% intervals of $\bar{\lambda}_1$ in Tables II and III describe the dispersion of λ_1 over each chaotic region and are positive by construction (Section III-D). The classification itself is supported by two checks. First, 91.8% of the 35,623 chaotic points of the 124 sweeps have an ensemble mean of λ_1 above two standard errors (from 87.9% to 98.3% in the configurations of Table II). Second, N_c depends on the threshold: raising it from 0 to 10^{-3} and to 5×10^{-3} keeps 73.1% and 56.9% of the chaotic points, respectively, so many chaotic points have small positive λ_1 and lie near the chaotic–periodic boundary. The dispersion σ_{λ_1} is highest for the rectangular bandstop filter (0.0491), whose fragmented chaotic region mixes low and high values of λ_1, and smaller for the lowpass filter with the Kaiser window (0.0231), whose chaotic region is more homogeneous. Table III ranks the six fixed windows and the best Kaiser window of each filter type by N_c.

The first six positions of Table III are lowpass filters, which confirms this type as the one of largest absolute chaotic coverage, and highpass and bandstop filters occupy mostly the lower half of the table. The bandstop configurations, although low in N_c, have the highest $\bar{\lambda}_1$ (up to 0.0929): when chaotic, their orbits separate faster than those of the lowpass filters. Whether this brings any benefit for communication was not evaluated here.

G. Computational Cost

The 124 sweeps comprise 372,000 grid points. The effective accumulation horizons recorded per grid point add up to about 2×10^8 iterations, that is, about 5×10^9 iterations of the expanded map once the 25 initial conditions of each point are counted. On an 8th-generation Intel Core i7-8550U processor (4 physical cores, 8 threads, 32 GB of RAM), the complete set of sweeps took 489 s, about 8 min, measured end to end. Any figure or table of this paper can be reproduced by running the corresponding script over the versioned sweep files (Section III-E), without recomputing the sweeps.

V. Final Remarks

This work investigated numerically how the filter type and the design window affect the preservation of chaos in the Hénon map with an FIR filter in the feedback loop. Four filter types (lowpass, highpass, bandpass, and bandstop) and seven windows (rectangular, Bartlett, Hamming, Hann, Blackman, Blackman–Harris, and Kaiser, with a sweep of its shape parameter β_K) were combined in 124 sweeps of the order–cutoff plane.

The insertion of the filter can change the generated signals from chaotic to periodic or divergent, depending on the pair (N_z, ω_c). The chaotic regime is not eliminated by filtering, but reorganized into regions of the parameter space whose shape depends on the filter type and the window. For lowpass and bandpass filters, windows of high attenuation (Kaiser with high β_K and Blackman–Harris) give the largest chaotic coverage, with $\rho_c^{\text{abs}} = 24.5\%$ for the lowpass filter with the Kaiser window, the highest among all configurations. For the bandstop filter, the rectangular window gives the highest conditional coverage among its configurations ($\rho_c^{\text{fin}} = 65.2\%$), although its absolute coverage is only 5.9%. The divergent fraction of the grid ranges from 6.77% (highpass) and 10.32% (bandpass) to 68.87% (lowpass) and 92.53% (bandstop). For $N_z \geq 8$, chaos is most frequent at low cutoff frequencies ($\omega_c/\pi \leq 0.1$), and the choice of window matters mainly at low and moderate orders.

The work makes three contributions. First, it extends the previous studies of the filtered Hénon map [14], [16], [17], in which the filters designed by windowing were lowpass filters with the Hamming or the Blackman window, to four filter types and seven design windows, with classification maps that show where chaos is preserved in each case. Second, it assesses the robustness of the classification with the ensemble standard error of λ_1, the sensitivity of N_c to the classification threshold, and repeated runs with ten seeds, which extend the point estimates reported in those studies. Third, it releases an open-source package [24] with which all the maps and tables can be reproduced and the study extended.

Taken together, the results show that the joint choice of filter type and design window is an important design factor of the filtered Hénon generator, especially at low and moderate orders, where the six fixed windows disagree on the class of up to 75% of the lowpass grid points. Because of the finely interleaved regime boundaries and of the shrimps inside the chaotic regions [18], the design of a filtered generator for CBCS cannot start from the spectral specification of the filter alone: the dynamics of the resulting generator must be verified explicitly, for example with classification maps such as the ones presented here.

The study has limitations. It covers the filter order, the cutoff frequency, the filter type, and the design window; other dimensions of the parameter space were not swept. The sweeps are limited to $N_z \leq 40$ and $N_{\text{map}} \leq 3{,}000$ iterations, and higher orders or longer horizons could reveal other structures. The band width $\Delta\omega = 0.05\,\pi$ of the bandpass and bandstop filters was fixed, and its variation may change the conclusions for these two types. The map parameters $(\alpha, \beta) = (1.4, 0.3)$

TABLE III
RANKING OF THE WINDOW–FILTER COMBINATIONS (FIXED WINDOWS AND BEST KAISER PER FILTER TYPE), BY N_c

#	Filter	Window	N_c	ρ_c^{fin}	$\bar{\lambda}_1$	95% CI
1	Lowpass	Kaiser (β_K=13.5)	979	77.6%	0.0239	[0.0223, 0.0254]
2	Lowpass	Blackman–Harris	961	74.8%	0.0211	[0.0197, 0.0227]
3	Lowpass	Blackman	801	61.8%	0.0176	[0.0160, 0.0193]
4	Lowpass	Hann	606	51.2%	0.0190	[0.0170, 0.0208]
5	Lowpass	Bartlett	458	41.3%	0.0198	[0.0174, 0.0226]
6	Lowpass	Hamming	450	35.9%	0.0263	[0.0240, 0.0290]
7	Bandpass	Kaiser (β_K=13.5)	434	13.0%	0.0286	[0.0258, 0.0315]
8	Bandpass	Blackman–Harris	418	12.3%	0.0261	[0.0235, 0.0293]
9	Bandpass	Blackman	394	11.1%	0.0252	[0.0218, 0.0290]
10	Bandpass	Hamming	294	7.9%	0.0273	[0.0235, 0.0315]
11	Bandpass	Hann	288	8.0%	0.0267	[0.0231, 0.0311]
12	Bandpass	Bartlett	249	6.9%	0.0293	[0.0254, 0.0337]
13	Lowpass	Rectangular	204	19.5%	0.0354	[0.0314, 0.0396]
14	Highpass	Kaiser (β_K=11.5)	149	8.1%	0.0178	[0.0138, 0.0219]
15	Highpass	Blackman–Harris	144	7.9%	0.0190	[0.0151, 0.0237]
16	Bandpass	Rectangular	141	3.7%	0.0153	[0.0113, 0.0211]
17	Bandstop	Rectangular	118	65.2%	0.0709	[0.0627, 0.0800]
18	Highpass	Blackman	97	5.2%	0.0216	[0.0170, 0.0277]
19	Highpass	Bartlett	86	4.6%	0.0237	[0.0186, 0.0299]
20	Highpass	Hann	79	4.2%	0.0233	[0.0173, 0.0310]
21	Highpass	Hamming	72	3.8%	0.0234	[0.0175, 0.0300]
22	Bandstop	Kaiser (β_K=4.0)	58	39.2%	0.0726	[0.0626, 0.0798]
23	Bandstop	Hamming	55	35.5%	0.0874	[0.0854, 0.0892]
24	Bandstop	Blackman	50	33.3%	0.0888	[0.0865, 0.0912]
25	Bandstop	Hann	49	32.9%	0.0877	[0.0860, 0.0896]
26	Bandstop	Blackman–Harris	47	32.0%	0.0929	[0.0909, 0.0946]
27	Bandstop	Bartlett	43	30.1%	0.0896	[0.0882, 0.0911]
28	Highpass	Rectangular	25	1.3%	0.0269	[0.0171, 0.0398]

95% intervals obtained by bootstrap [30]. Full per-configuration tables are regenerated from the versioned sweep files by the released package [24].

were not swept, and other pairs of the parameter space of the Hénon map [18] may behave differently under filtering. Finally, initial conditions that diverge after the burn-in are excluded from the mean of λ_1, which is therefore estimated over the surviving initial conditions.

Three directions are proposed for future work: (a) sweeping the band width $\Delta\omega$ as an independent variable for bandpass and bandstop filters, to identify transitions not captured in this study; (b) extending the methodology to IIR filters, whose dynamic behavior is still little explored in the literature [19], with explicit treatment of the stability of the filter itself; and (c) measuring the occupied bandwidth, the spectral leakage, and the in-band power of the chaotic configurations identified, as metrics complementary to λ_1 for the choice of operating points in CBCS.

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